SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –SIH26028**
- **Problem Statement Title-Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains**
- **Theme-Software Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal):Rail time Ai**



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IDEA TITLE : AI BASED SMART BLOCK PLANNING

Maximizing Asset Availability for Train Operations

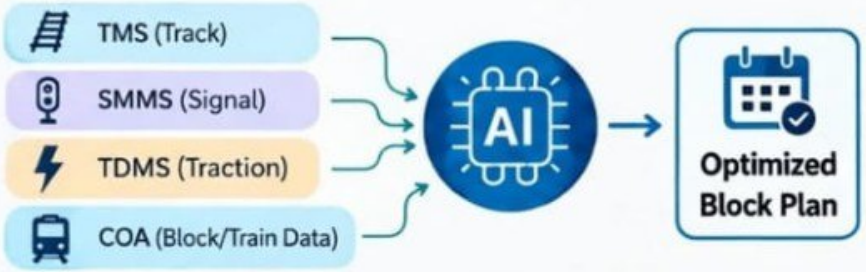
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The Problem

- Maintenance is planned separately by different departments (Engineering, Traction, Signal & Telecom).
- Data is scattered across multiple systems (TMS, SMMS, TDMS, COA).
- Manual and decentralized planning leads to:
  - Poor coordination
  - Overlapping/extra blocks
  - Increased asset downtime
  - Impact on train operations

Our Idea

An **AI-powered system** that integrates maintenance, defect and train operation data to automatically generate optimized block schedules, ensuring maximum asset availability and minimal impact on train operations.



Key Features

- Real-time integration of data from TMS, SMMS, TDMS & COA
- AI/ML based prioritization of maintenance tasks (criticality, urgency, safety, condition)
- Smart block scheduling with train timetable & goods train forecast
- Multi-department coordination (avoid duplicate blocks)
- Weekly & monthly block plans (short + long term)

How It Addresses the Problem

Current Challenge		Our Solution
Departments plan blocks separately	→	One centralized system
Manual scheduling	→	AI-based automatic scheduling
Overlapping / repeated blocks	→	Combine compatible tasks
Maintenance affects train operations	→	Selects slots with minimum train impact
Critical tasks may be delayed	→	AI prioritizes urgent & critical tasks
More asset downtime	→	Optimizes block duration to maximize availability

From Multiple Systems → To One Smart Plan



Innovation & Uniqueness

One Platform → Smart Priority → Coordinated Block → Minimum Downtime → Maximum Asset Availability

- Brings all departments together instead of isolated planning.
- Uses AI to intelligently combine tasks and find the best common block time.
- Balances maintenance urgency with train operation needs.
- Creates a safer, faster and more efficient railway network.

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TECHNICAL APPROACH

AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways

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TECHNOLOGIES & TOOLS

Frontend: React / Flutter
Interactive **dashboard** for **railway officials**

Backend: Python + FastAPI / Node.js
Process and manage block requests

AI/ML: Python, Scikit-learn + optimization algorithms
Smart block scheduling

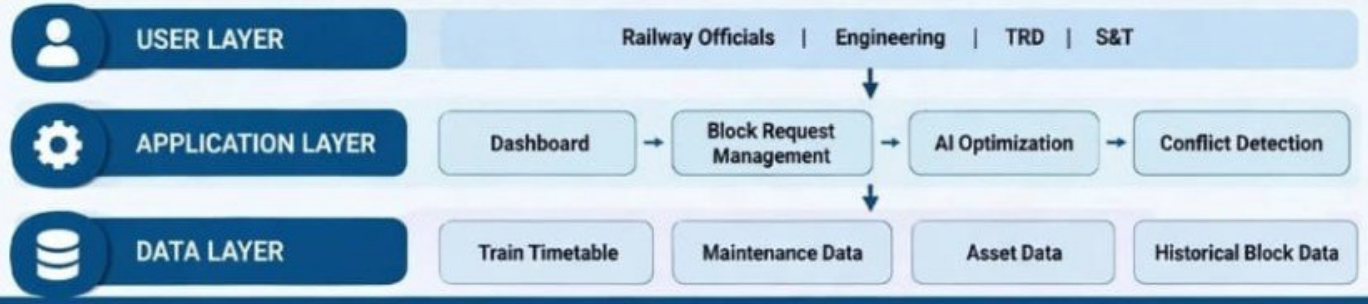
Database: MySQL / PostgreSQL
Train schedules, assets and maintenance requests

Visualization: Charts + Timeline View
Available maintenance windows

SMART BLOCK PLANNING FLOW



3-LAYER ARCHITECTURE



IMPLEMENTATION STATUS

- AI Scheduling Engine
- Conflict Detection
- Interactive Dashboard

Prototype architecture & workflow

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FEASIBILITY AND VIABILITY

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A practical solution • Realistic implementation • Long-term impact

ANALYSIS OF THE FEASIBILITY OF THE IDEA

Technical
Uses existing railway data and AI for smart block planning.

Financial
Software-based solution that can reduce delays and inefficient block usage.

Market
Can be scaled across railway divisions and other transport systems.

Operational
Improves coordination between different maintenance departments.

POTENTIAL CHALLENGES AND RISKS

Financial Risks
- Initial development and integration cost.
- May increase maintenance cost during large-scale implementation.

Market Risks
- Slow adoption of a new AI-based system.
- Need to prove better results than manual planning.

Operational Risks
- Data from different departments may not be properly connected.
- Sudden delays or emergency maintenance can affect the plan.
- Staff may need training to use the system.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

Methods
- Start with a pilot project.
- Test using real and past railway data.

Principles
- Keep the system simple, reliable and easy to use.
- Minimum disruption to train services.

Strategies
- Gradual implementation.
- Training and support for railway staff.

Algorithms
- AI prodiction for maintenance needs and delays.
- Optimization for best block time.
- Dynamic rescheduling for real-time changes.

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IMPACT AND BENEFITS

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Smarter Planning | Better Coordination | Stronger Railways

POTENTIAL IMPACT ON TARGET AUDIENCE



Improvement

- Better coordination between maintenance and train operations.
- Reduced block conflicts and train disruption.

Economic Impact

- Better use of time and resources.
- Reduced unnecessary costs and operational delays.

New Opportunities

- Supports AI-based smart railway planning.
- Can be expanded to other railway divisions in the future.

Social Benefits

- More reliable train services for passengers.
- Better decision support for railway staff.

BENEFITS OF THE SOLUTION

Social

Improved Access	• More availability of railway services.
Improvement	• More reliable and smoother train operations.
Reduction	• Fewer delays and less manual coordination work.

Economic

Productivity	• Faster and smarter maintenance planning.
Cost	• Better resource use and reduced operational losses.
Market	• Can be scaled across railway divisions and other transport networks.

Environmental

Energy	• Reduces unnecessary train waiting and energy use.
Reduction	• Less disruption and inefficient train movement.
Waste	• Better planning helps avoid wasted maintenance time and resources.

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RESEARCH AND REFERENCES

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1. EXISTING APPS / SYSTEMS STUDIED

a) NTES – National Train Enquiry System

- Live train running status
- Expected arrival/departure
- Delay and station information

CRIS – NTES

b) RailOne – Indian Railways passenger super app

- Train status
- PNR enquiry
- Ticketing and passenger services

Indian Railways – RailOne

c) RTIS – Real-Time Train Information System

- GPS-based real-time train/locomotive movement data
- ETA improvement potential

Indian Railways – RTIS

d) COA – Control Office Application

- Train movement events
- Control-office monitoring
- Updates for passenger information

CRIS – COA

2. RAILWAY OPERATIONAL INPUTS STUDIED

a) Railway Signalling

- Signal aspects
- Block/section status
- Operational constraints

Indian Railways Signal & Telecom

b) Railway Electrification / OHE

- Traction power/electrification status
- Related operational conditions

Indian Railways Electrification/OHE

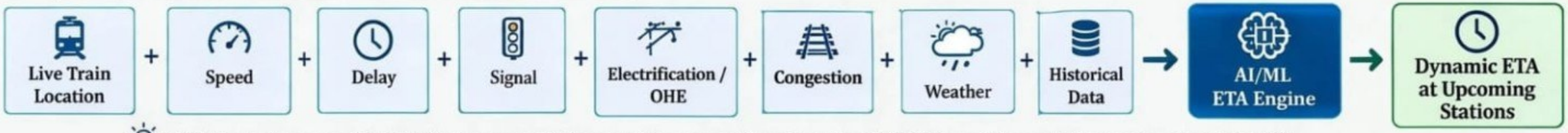
c) Track & Route Conditions

- Congestion
- Speed restrictions
- Maintenance blocks
- Unscheduled stoppages

d) Weather & Historical Data

- Weather conditions
- Historical delay/running patterns

3. OUR PROPOSED APP – DYNAMIC ETA



Existing systems provide data/status; our solution combines operational inputs with ML to continuously forecast and update ETA.

THANK YOU

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